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HOMOGENEITY ADJUSTMENTS OF TEMPERATURE AND PRECIPITATION SERIES—FINNISH AND NORDIC DATA

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ABSTRACT

An analysis is made of the adjustments needed to produce three homogeneous data sets, namely the 1961–1990 mean temperatures in Finland, the North Atlantic Climatolological Dataset (NACD) temperature and precipitation series (1890–1990), and the Finnish daily mean maximum and minimum temperature series (1910–1995), as well as the reasons for making such adjustments. The adjustments in the annual (seasonal) mean temperatures are up to ± 1°C (± 2°C), and annual precipitation adjustments can be ± 40%. In Finland, the homogeneity breaks in the normal period temperatures and in the long-term daily mean maximum and minimum temperatures appear to be random, and thus, do not bias averages based on large numbers of stations. However, both the temperature and precipitation series of the NACD would have been statistically significantly biased without adjustments. Station relocations appear to be the most common cause of homogeneity breaks in the temperature series. In the NACD, the adjustments resulting from relocations are statistically significant and reflect changes to colder observing sites. Also, changes in the formula used for the calculation of mean temperatures and urbanization both cause systematic biases in the data. The installation of improved precipitation gauges has been systematic in the NACD; thus, the original series need to be adjusted upwards in the early years. The applied adjustments are of the same order of magnitude as the observed long-term trends, which stresses the importance of the testing and adjusting of long-term series before analysis of climatic changes. In order to monitor climatic changes in a reliable manner, the observing network should be designed to withstand the common discontinuities (e.g. relocations, observer and environment changes etc.) in observation series, because the number of homogeneity breaks appears to be roughly constant in time. Moreover, the introduction of new technology may cause systematic changes in the observations, and comprehensive comparison measurements are needed. Copyright © 2001 Royal Meteorological Society.

KEY WORDS: homogeneity testing; metadata; Nordic; precipitation; temperature

1. INTRODUCTION

Long climatological time series are often plagued with artificial discontinuities, for example, those caused by station relocations, changes in instrumentation and changes in observing practices. Several types of disturbances can distort, or even hide, the true climatic signal. It is, therefore, quite natural that climatological time series are tested in order to detect possible inhomogeneities. There are several statistical methodologies for homogeneity testing and adjusting. Peterson *et al.* (1998) present a review of methods. All climatic studies and data sets should be based on homogeneity-adjusted data because the original time series could give a misleading description of the evolution of climate.

Some articles have focused on the adjustments to specific inhomogeneity sources, for example, Karl *et al.* (1986) on time-of-observation bias, Parker (1994) and Nordli *et al.* (1997) on radiation screens, and Jones *et al.* (1990) and Portman (1993) on urbanization. However, there have not been many attempts to analyse quantitatively all the inhomogeneity sources of complete data sets. A notable exception is a study by Hanssen-Bauer and Førland (1994), who analysed all the adjustments needed to produce a

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homogeneous data set for a Norwegian precipitation time series. They even studied available metadata, and attributed the homogeneity breaks to various different causes. For global or hemispheric data sets, it is not possible to perform such an analysis simply because the necessary metadata is not available.

An analysis of the adjustments used to create homogeneous data sets enables one to tackle many interesting questions. How large are the homogeneity breaks? How common are they? What kinds of breaks are systematic? Can analysis of past homogeneity breaks give guidance in maintaining an observing network?

This paper reviews the size, timing, and causes for adjustments in Finnish and Nordic temperature and precipitation series. Homogeneity tests or comparison measurements have been used to determine the size and timing of the breaks. The causes of the homogeneity breaks have been sought from the metadata (meteorological yearbooks, inspection reports etc.). However, because the station history information is not complete, it is not always possible to discover the cause(s) of the discontinuities.

The adjustments of three data sets are reviewed. First, the Finnish annual mean temperatures of the standard normal period 1961–1990 have been tested and adjusted. Second, during the North Atlantic Climatological Dataset (NACD) project, the long-term mean temperatures and precipitation totals have been homogenized (Frich *et al.*, 1996). Third, Finnish mean daily maximum and minimum temperatures have been tested for the Relating Extreme Weather to Atmospheric circulation using a Regionalized Dataset (REWARD) project (Førland *et al.*, 1998). The testing practices are briefly described and the homogeneity breaks analysed. In conclusion, some discussion on the influence of adjustments is presented.

2. NORMAL PERIOD MEAN TEMPERATURES

Normal period values are often used as a yardstick in climate monitoring. For example, monthly mean temperatures are presented together with their respective anomalies from the standard normal period. Any problems with homogeneity during the normal period of 1961–1990 are reflected in the monthly climate bulletins, which are routinely published by many of the National Meteorological Services (NMS). In Sweden, the temperature and precipitation normal values have been tested and adjusted (Alexandersson *et al.*, 1991). In Finland, and many other countries, normal values have been published without homogeneity testing. In this study, the normal period mean temperatures have been tested.

2.1. Description of the test method and data

Homogeneity testing was performed with the Standard Normal Homogeneity Test (SNHT) developed by Alexandersson (1986). This version of SNHT searches for a single shift, step-like discontinuity, relative to the reference series. SNHT compares the station being tested with homogeneous reference stations (four to nine stations during the normal period). In practice, it may be difficult to determine beforehand which stations are homogeneous. Testing is, therefore, usually done as an iterative procedure in which the homogeneity of the data is gradually improved by testing all series several times (Hanssen-Bauer and Førland, 1994; Tuomenvirta and Drebs, 1994; Moberg and Alexandersson, 1997).

All annual mean temperature series that consisted of more than 15 years of observations during the period 1961–1990 were tested. In order to detect breaks near the ends of the normal period also, the testing was performed for the period 1958–1993, if data were available. However, breaks at the beginning of the normal period, during the first few years or so, have only minor effects on the normal period mean value, which is fixed to the station characteristics in 1990.

The testing was performed on the annual mean temperatures, and the analyses of annual adjustments are presented. The exact timing and cause of the discontinuities were sought from station history information. SNHT discovered several breaks where no explanation from history documents could be found. Metadata is based on several sources, such as station inspections, the list of observers, list of instruments etc., but it is not collected comprehensively. It can, therefore, be assumed that most of the unexplained homogeneity breaks are real, even though no information about their cause can be found

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from the archives. In the final stage, all homogeneity breaks exceeding the 95% significance level were adjusted. The use of a new homogeneity testing method for the detection of linear trends by Alexandersson and Moberg (1997) and testing of seasonal instead of annual series (Moberg and Alexandersson, 1997) might have detected even more breaks than the methodology applied in this study.

2.2. Statistics of the adjustments

In total, 148 Finnish mean temperature series were tested. During the normal period, the station network is dense enough for efficient homogeneity testing. Typical correlation coefficients of the annual mean temperature between candidate and reference stations range from 0.93 to 0.99. This enables the detection of relatively small breaks. If stations were equally distributed, the average distance between stations would be about 50 km. It turns out that the homogeneous, as well as the non-homogeneous stations are spatially randomly distributed over Finland. It was found that rather more than one third of the time series contained relative homogeneity breaks (Table I). Exclusion of non-homogeneous stations would have created a coarse resolution data set. Therefore, the non-homogeneous series were adjusted.

The total number of breaks is greater than the number of non-homogeneous stations (Table I) because some series contained multiple breaks. The adjustment is defined as homogeneous values minus original values. Statistically, according to the Student's t-test (two-tailed), the mean of annual adjustments is not significantly different from zero. Moreover, monthly adjustments are below the 1% significance level. This supports the hypothesis that there have not been any systematic changes biasing the Finnish station network during the normal period.

The standard deviation (S.D.) of annual mean temperatures in Finland is typically about 1.1°C (Heino, 1994). Theoretically, if the largest break in Table I, $+0.48$°C, occurred in 1990, it would lead to a 13% overestimation of the above-normal years, and an 8% underestimation of the below-normal years in climate monitoring (the normal being defined as ± 1 S.D. from the mean, assuming a normal distribution). Similarly, an average-sized adjustment (average of adjustments) placed in the middle of the period 1961–1990 causes a 0.11°C bias in the normal period annual mean temperature, and a 2.5/2.3% error in the classification of the above/below normal years.

In reality, the differences between the published (FMI, 1991) and the homogenized normal period annual mean temperatures at stations that had experienced a homogeneity break are generally less than 0.2°C. The largest difference is 0.4°C. On a monthly level, the largest differences are up to 1.0°C. However, these errors are not only caused by homogeneity breaks, but also by the improper handling of years with missing data. Thus, the use of the original climate normals in monitoring is not a large problem, but it can bias some individual stations. Furthermore, because the observing network is continuously changing and homogeneity breaks occur, the normal period mean values 'get old', and no longer serve the purpose of being a reliable yardstick in monitoring.

During the period 1961–1990, the observing methods have remained practically unchanged. Therefore, almost all of the explained homogeneity breaks are caused by station relocations. Even small site changes,

Table I. Test results of annual mean temperatures in Finland, 1961–1990

	Homogeneous	Non-homogeneous	
		Breaks explained by station history	Not explained
Number of stations	94 (63.5%)	34 (23.0%)	20 (13.5%)
Number of breaks		39	20
Annual adjustments			
Mean		−0.00°C	0.05°C
S.D.		0.26°C	0.18°C
Largest positive		0.48°C	0.43°C
Largest negative		−0.49°C	−0.19°C

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by 10 m or so, could cause a discontinuity in the temperature series. Changes in the environment of the measuring site cause inhomogeneities on the local scale. These changes are real changes in the local climate, and are, therefore, not included in Table I. For example, deformation of peatlands during the 1960s and the 1970s has affected climate at some sites in Finland (Venäläinen *et al*., 1999). No adjustments should be made, if the local evolution of climate is to be studied. On the other hand, if a station that has experienced local environmental changes is used to describe climate of a large area, adjustments may be needed.

3. THE NACD TEMPERATURE AND PRECIPITATION SERIES

One of the objectives of the NACD project was to produce a consistent climatological data set covering the North Atlantic region during the period 1890–1990 (Frich *et al*., 1996). The data set is constructed from high-quality time series of monthly mean temperatures and precipitation totals.

From Finland, Norway, Sweden and the UK (stations in Scotland), it was possible to obtain information on the size, date, and causes of adjustments. Data from these countries make up 70% of the temperature and precipitation series in NACD. Adjustments were calculated from comparison measurements, whenever they were available. In Finland, Norway, Sweden and the UK, SNHT was further used to evaluate the homogeneity of the data (Frich *et al*., 1996). In temperature series, additive adjustments (differences) were used, whereas multiplier adjustments (ratios) were used for precipitation series.

3.1. The NACD series compared with the original data

In total, 57 temperature series and 72 precipitation series described as 'tested' or 'homogeneous' in the NACD catalogue (Frich *et al*., 1996) were included in the analysis. Ten (18%) of the temperature series and 17 (24%) of the precipitation series were homogeneous without adjustments. The rest of the temperature and precipitation series (47 (82%) and 55 (76%), respectively) required adjustment.

The annual adjustments (the homogenized value minus the original value) are displayed in Figure 1, so that the influence of adjustments on the original series can be seen. Some of the adjustments are close to zero/one, if there are equal size, but opposite breaks in the time series. The adjustments are distributed fairly evenly over the whole time period 1890–1990. There are no statistically significant differences in the number of adjustments between the first and last half of the study period.

All Finnish temperature data have been adjusted in 1900 and 1926 owing to changes in the formula used for the calculation of the daily mean. All precipitation series have been adjusted in 1909 and 1981 as a result of nation-wide changes in the precipitation gauge type (Heino, 1994). These discontinuities would have been difficult to detect and adjust with SNHT (or with any relative homogeneity testing method), as all series were affected at the same time. Fortunately, comprehensive comparison measurements have been performed (Heino, 1994 and references therein).

Are the homogenized series different from the original data? One way to answer this is to use the Student's t-test to find out whether the mean temperature/precipitation adjustments are statistically different (the alternative hypothesis) from zero/one (the null hypothesis). The results are displayed for seasonal and annual means in Table II.

All adjustments in Table II are significant, at least at the 1% significance level, most of them even at the 0.1% level. Temperature series, especially, needed adjustments during the summer, and precipitation series in the winter. Without adjustments, the annual mean temperatures would have been systematically too high, and the annual precipitation sums too low.

Figure 2 describes the time evolution of the analysed network and adjustments. Close to 90/70% of the temperature/precipitation series cover the full period and, generally, the rest of the stations started operation during the first two decades (Figure 2(a)).

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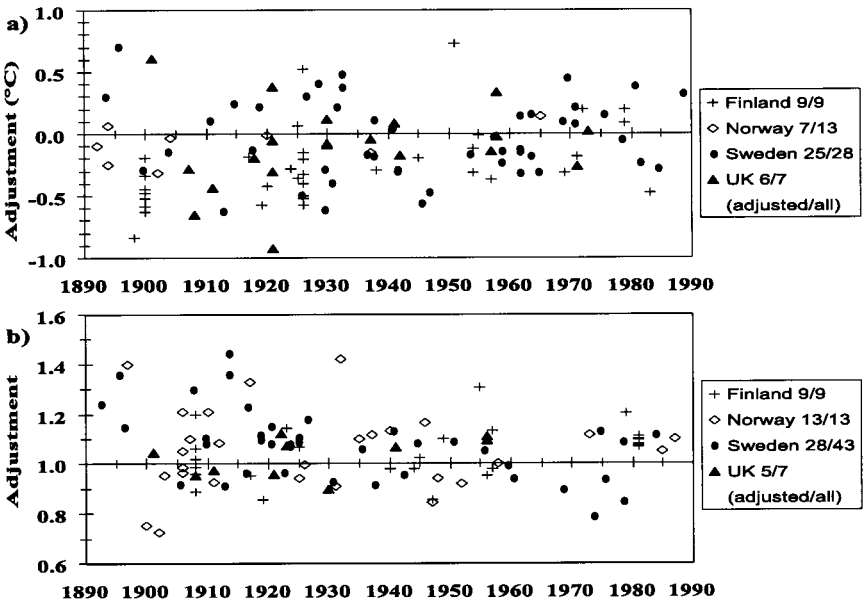


Figure 1. Adjustments used for the NACD (a) annual mean temperatures (additive) and (b) annual precipitation sums (multiplier) in Finland, Norway, Sweden and the UK. The number of adjusted/all stations is given after the country name

Table II. Mean seasonal and annual adjustments (homogeneous minus original value)

	n	DJF	MAM	JJA	SON	Year
Temperature (°C)	121	−0.09**	−0.10**	−0.22***	−0.10***	−0.13***
Precipitation	106	1.087***	1.065***	1.023**	1.039**	1.054***

Statistical significance is tested with a two-tailed Student's *t*-test (* 95%, ** 99%, *** 99.9%).
Column *n* gives the number of breaks.
DJF = December, January, February. MAM = March, April, May. JJA = June, July, August. SON = September, October, November.

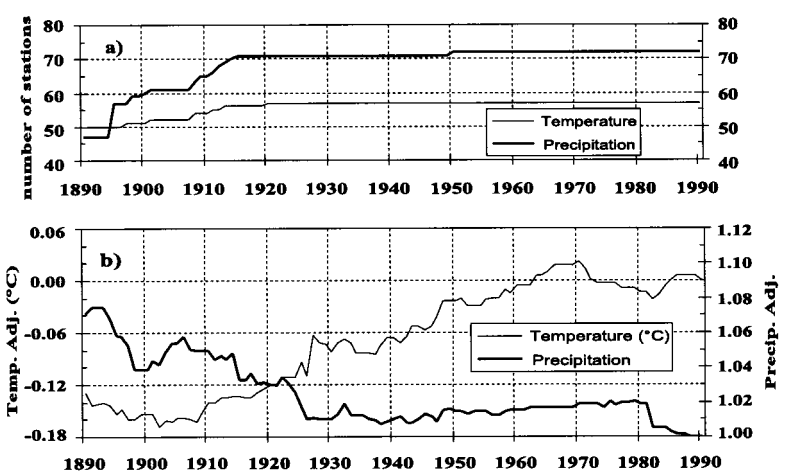


Figure 2. (a) The number of stations used in the analysis (Finland, Norway, Sweden and the UK). (b) The mean annual adjustments in the NACD series

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During the period 1960–1990, the mean of the temperature adjustments is close to zero (Figure 2(b)). Going further back in time, the temperature adjustments clearly become negative during the 1950s, and continue to become more negative until the beginning of the century. The precipitation adjustments 'rise' in 1981, but show a slight decline until the 1940s. The precipitation adjustments have a tendency to cancel each other, although more and more stations need adjustments. At the beginning of the 20th century, the adjustment factor is larger than 1.03. In the end of the 19th century, the original showed about 6% too little precipitation.

3.2. Reasons for adjustments

Another consideration regarding the adjustments is to classify them according to the reasons for the homogeneity breaks, as Hanssen-Bauer and Førland (1994) did for the Norwegian precipitation series. Reasons were sought from station history information. *Instrument changes* were defined as changes in radiation screen type or precipitation gauge type, including the installation of a windshield. In general, there has been a tendency to improve temperature measurements by increasing both the screening of thermometer and the ventilation with new designs. New gauges and windshields have been installed to alleviate the aerodynamic catch deficiency of precipitation gauges. *Relocation* of the temperature screen or precipitation gauge is the most common cause of a homogeneity break. Even a small site change may move the screen/gauge into a different temperature/wind microclimate. A *change in the environment* of the measuring site (e.g. urbanization, construction of buildings, changes in vegetation) may cause either abrupt or gradual discontinuities. Changes in observing practices or in the processing of data may involve a large number of stations, as in the case of Finnish mean temperatures, where *changes in the formula* used for the calculation of mean temperature caused homogeneity breaks.

The classification of the reason for a homogeneity break requires some subjective choices. When there are multiple changes at the same time, it is usually impossible to separate the different effects. For example, before the standardization of measurement heights, relocations of screens/gauges were sometimes accompanied with changes in the screen/gauge height. In this study, in the case of multiple changes, relocation has always been defined as the reason for adjustment, because relocations are known to cause frequent and large homogeneity breaks (Hanssen-Bauer and Førland, 1994 and the previous chapter).

Table III gives the number and mean sizes of the relative adjustments (difference or ratio between consecutive homogenous sub-periods). The old screen installations gave higher values than the modern screens. Nordli *et al.* (1997) analysed comparison measurements from different radiation screens and found that, in general, old wall screens and free-standing, single louvred screens are warmer than modern, double louvred screens during summer and spring, but during winter and autumn, the test results were

Table III. Seasonal and annual means of relative temperature and precipitation adjustments classified according to the reasons for homogeneity breaks

	n	DJF	MAM	JJA	SON	Year
Temperature (°C)						
Instrument/screen changes	20	−0.17**	−0.09	−0.14*	−0.11*	−0.13**
Relocations	43	−0.13*	−0.09	−0.11	−0.12*	−0.11*
Environment changes	6	0.19*	0.28**	0.22**	0.18**	0.22**
Formula changes	16	−0.04**	−0.19***	−0.40***	−0.03	−0.17***
Unknown reasons	36	−0.03	−0.04	−0.04	−0.02	−0.03
Precipitation						
Instrument changes	27	1.178***	1.112***	1.027**	1.080***	1.099***
Relocations	34	0.980	0.983	0.993	0.981	0.985
Environment changes	17	1.065	1.037	1.014	1.026	1.036
unknown reasons	28	1.095*	1.076**	1.043*	1.046*	1.065**

Statistical significance etc. as with Table II.

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inconclusive. In this study, autumn and winter adjustments are also significantly negative (old screens warmer). In a high-latitude, non-maritime climate, surface inversions are common during the winter. One possible explanation for wintertime negative adjustments is the lowering of the measurement height (not always documented) that is sometimes coincidental with changes from wall screens to free-standing screens.

The temperature screen relocations have occurred somewhat systematically towards cooler sites. It is unlikely that the breaks referred to here as 'relocations' are biased by screen changes performed in connection with the site changes, because only four combined changes (relocation and screen change) were documented out of 43 breaks. Rather, metadata suggests that during the 1930s, 1940s and 1950s, there has been a tendency to move stations from closed areas in growing towns to more open sites, for example, to airports. This can be seen as a counter-action to increasing urbanization.

Environment changes affecting temperature result mostly from urbanization. The adjustments are thus large and significantly positive. However, there are only six of them in the NACD. Homogeneity breaks caused by formula changes are systematically negative, indicating that the old methods of calculating daily mean temperatures in Finland were providing too high mean temperatures, especially during summer (Heino, 1994). The temperature adjustments caused by unknown reasons are, on average, small and random.

Normally, precipitation measurements underestimate the 'true' precipitation owing to wind error, evaporation, wetting error etc. (Sevruk, 1982). This problem can be reduced by instrument design, and by placing the precipitation gauge in a suitable environment. Instrument changes have decreased the measurement error, and the old values must be adjusted 'upward' (the adjustment factor is > 1) in winter by almost 18% (Table III). In particular, the introduction of windshields has reduced the deficit in measured solid precipitation resulting from wind error (Hanssen-Bauer and Førland, 1994).

Adjustments owing to relocations of precipitation gauges are consistently below unity from season to season (Table III), but they are not statistically significant. Environment changes produce opposite-directed, non-significant adjustments. Frequently, the physical explanation for breaks owing to relocation or environment change is a change in the exposure of gauge leading to a change in the gauge catch. Hanssen-Bauer and Førland (1994) point out that, in areas with large gradients in precipitation amount or in the ratio of solid and liquid precipitation (mountainous regions), relocations may place the measurement in a truly different precipitation climate. Surprisingly, breaks for unknown reasons give systematically positive changes, that is, appear to give adjustments significantly larger than one. One possible explanation is that this category contains undocumented instrument changes, and/or decreased wind exposure owing to undocumented vegetation growth and/or the construction of new buildings.

4. FINNISH MAXIMUM AND MINIMUM TEMPERATURES

As part of the REWARD data set, monthly mean daily maximum and minimum temperatures were collected for an analysis of extremes. In Finland, a large set of stations was tested to produce 19 homogeneous time series of monthly mean daily maximum and minimum temperatures. The Finnish data were part of those utilized by Tuomenvirta *et al.* (2000) in the analyses of maximum and minimum temperatures, and diurnal temperature range (DTR) in the Nordic region.

4.1. Test results

Applying SNHT to seasonal series, and using station history information, 93/90 maximum/minimum temperature series were tested. The aim was to create a subset of good quality series, not to homogenize all the data. Therefore, in the early phase of testing, some stations were already excluded if they were inhomogenous and were not necessary in the homogenization of REWARD stations (Table IV). Homogenization was continued with 63/60 maximum/minimum temperature stations. Some of the series were fairly short (even less than 30 years) compared with the Finnish REWARD temperature series

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Table IV. Test results of the annual mean daily maximum and minimum temperature series in Finland, 1901–1995

Stations (number/percentage)	Homogeneous	Non-homogeneous	
		Adjusted	Excluded
Maximum	30/32	33/36	30/32
Minimum	23/26	37/41	30/33

(average length 87 years). Some of the stations no longer operate, but proved to be useful in the homogenization process, where from two to nine reference stations were used for testing the candidate station.

Table V contains the mean sizes of seasonal and annual relative adjustments (difference between consecutive homogenous sub-periods). On average, the adjustments of minimum temperatures are more negative than the adjustments of maximum temperatures, and there are more of them. All seasonal mean adjustments are negative, indicating that the old, original values were too high. However, only the wintertime minimum adjustments are significantly different from zero.

According to the metadata, station relocations and screen changes are by far the most common causes of discontinuities. In summer, the adjustments of maximum temperatures are usually larger than in winter (Figure 3(a)). During summer, the albedo of the screen and/or the environment, as well as the ventilation, strongly influence maximum temperatures, leading to large differences between different sites and screens. During winter, the albedo and ventilation have a much smaller effect.

In Finland, the transition to modern screens mainly took place in the 1910s and 1920s, when most of the maximum temperature adjustments are negative (Figure 3(a)). Although the sign of the adjustments agrees with the comparison measurements reported in Nordli *et al*. (1997), it must be kept in mind that the screen changes often took place in connection with site changes, and it is not possible to separate their effects.

The monthly mean daily minimum temperatures have large spatial variability because local topography and heat sources/sinks (e.g. lakes) have an effect on the minimum temperatures. Relocations may, therefore, cause large adjustments (Figure 3(b)). The largest adjustments of monthly temperatures (not shown) were as high as 2.2°C for the minimum, compared with the largest monthly maximum adjustment of 1.6°C.

The minimum temperature adjustments during the 1940s and 1950s are mostly negative (Figure 3(b)). This is a result of somewhat systematic relocations in the station network. During those decades, eight city centre stations were moved to airports; other moves out of town centres also took place. Open airport sites are sensitive to radiative cooling, and negative adjustments for minimum temperatures are thus needed. Otherwise, both maximum and minimum temperature adjustments are evenly distributed during the period 1901–1995.

Table V. Seasonal and annual means of relative maximum and minimum temperature adjustments

	n	DJF	MAM	JJA	SON	Year
Maximum (°C)	59	−0.04	−0.05	−0.07	−0.06	−0.06
Minimum (°C)	73	−0.14*	−0.07	−0.11	−0.08	−0.10

Statistical significance etc. as with Table II.

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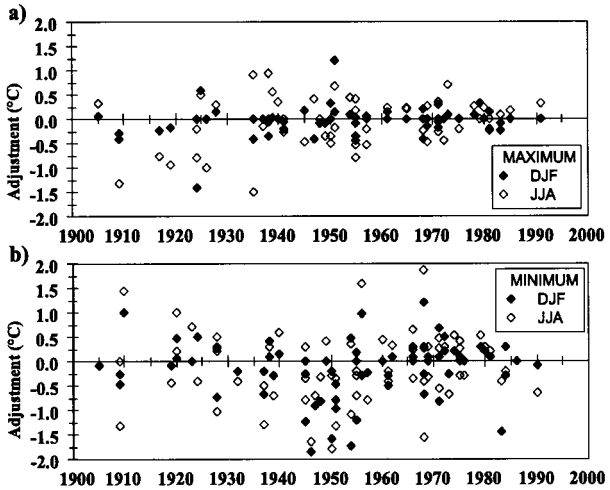


Figure 3. The relative adjustments used for (a) mean daily maximum temperatures and (b) mean daily minimum temperatures in the Finnish winter (December, January, February) and summer (June, July, August)

4.2. Comparison of mean and extreme temperature adjustments

It is interesting to compare the adjustments of the monthly mean (calculated from the fixed-hour measurements with a dry-bulb mercury thermometer), the mean daily maximum (measured with a maximum thermometer), and the mean daily minimum temperatures (measured with a minimum thermometer). This can be done for 18 REWARD stations in Finland during the period 1901–1990.

The daily maximum and minimum temperatures are more sensitive to the screen construction, the measurement height and the environment than are the mean temperatures. There are roughly 30% more discontinuities in the maximum and minimum temperature series than in the series of mean temperatures, if adjustments caused by formula changes are not counted. However, if the adjustments caused by changes in the calculation method of daily mean temperature are counted, the mean temperature series have homogeneity breaks slightly more often than do the extreme temperatures. This is a characteristic of Finnish data. In Sweden, a switch of formulae for monthly averages in 1900–1901 had no effect on the mean level (Moberg and Alexandersson, 1997). In Norway, Nordli (1997) found changes in observation time and processing that affected mean daily minimum temperatures.

5. DISCUSSION

On average, the adjustments made to Finnish normal period mean temperatures, monthly mean daily maximum and minimum temperatures are not significant. However, adjustments for individual stations can be quite large. Besides single stations, area-averages based on a small number of stations also can be biased. For example, in Figure 4, the long-term springtime temperature increase in southwestern Finland would have been underestimated without homogeneity testing and adjusting. It can also be seen how the uncertainty is reduced.

Tuomenvirta *et al.* (2000) report a statistically significant decrease in the DTR over Fenno-Scandia during the period 1910–1995. They link it to the increase in cloud cover. Although the observed DTR trends are significant, they are rather small, and could not be considered reliable without proper testing and adjustment of the maximum and minimum temperature series.

The NACD would not have contained many series if the originally homogeneous data only had been accepted. Adjustment was also essential because there was a systematic bias in the original data. The old mean temperatures were too high and the old precipitation sums too low. The magnitudes of the mean

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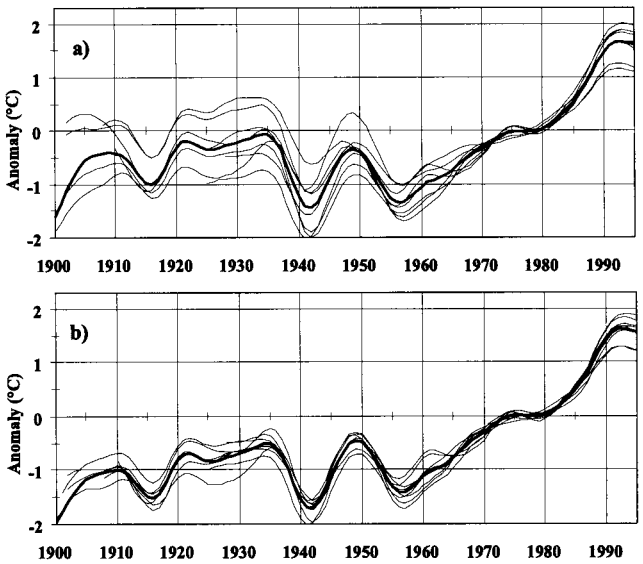


Figure 4. Smoothed spring (March, April, May) anomalies of (a) original and (b) homogenized mean daily minimum temperatures in 1900–1995 (reference period 1961–1990). The thick curve is the mean of eight stations located in southwestern Finland (inside the latitude–longitude box 60°–65°N, 20°–25°E)

annual adjustments are 20–60% of the observed positive temperature trends (0.2°–0.7°C/100 years) within the NACD region (Hanssen-Bauer *et al*., 1996). The annual precipitation adjustments are also comparable with the observed trends (3–14%/100 years) (Hanssen-Bauer and Førland, 1996; Tuomenvirta and Heino, 1996). Without homogenization, it is not possible to make reliable estimates of the long-term climate changes.

The mean duration of homogeneous sub-periods turned out to be about 35 years for the long mean, maximum and minimum temperature series in Finland. Auer and Boehm (1996) report a value of 27 years based on long-term Austrian temperature series. The probability of a series classified as homogeneous increases as the period being tested becomes shorter. The homogeneity testing of the period 1961–1990 annual mean temperatures in Finland gives the mean duration of homogeneous sub-periods of about 60 years.

The findings of this study may not hold for older data, or for other countries. Nevertheless, studies made in Austria by Auer *et al*. (1999) show many similar homogeneity problems to those found in this study. NMS often have expertise in the history of observational practices, which should be utilized when long-term data sets are constructed.

Concerning the maintenance of observation networks, two lessons can be learned from the analysis of past homogeneity breaks. First, recent decades have as many homogeneity breaks as earlier ones. This suggests that there has not been any improvement in terms of homogeneity. Climatological observation networks should be designed to withstand relocations, observer changes and other often, but not always, random-like discontinuities, which will eventually affect most stations. Second, from time to time, major changes will be made in methods of observing, which may significantly bias the whole network. For example, the International Meteorological Congress in Vienna in 1873 started the international standardization of meteorological observations. This caused nation-wide homogeneity breaks during the following decades when NMS revised their observing practices. At the present time, the increasing introduction of new technology, for example, automatization and satellite observations, despite its advantages, is a potential threat to the homogeneity of surface climatological observations if comprehensive comparison measurements are neglected.

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6. CONCLUSIONS

The main conclusions are as follows.

- The overall influence of homogeneity breaks on the normal period mean temperatures in Finland was small and random. However, the normal values of individual stations had errors of up to 0.4°/1.0°C in the published annual/monthly values owing to homogeneity breaks and improper handling of missing data.
- Long (> 30 years) time series are often inhomogenous and need adjustments before climate change analyses. Inhomogeneities can be large, and can hide the climatic signal in individual time series.
- In Finland, extreme temperatures appear to be more sensitive to relocations, screen changes, and environment changes than mean temperatures, which are calculated from fixed hour observations. However, the mean temperatures suffer from methodological changes.
- Both temperature and precipitation series contain systematic errors that can bias even large-scale area-averages. Changes in instruments and observing practices, urbanization and, rather unexpectedly, station relocations caused significant systematic biases in the original NACD temperature series. The NACD precipitation series were systematically biased by instrument changes. In the Nordic region, the mean magnitudes of the adjustments are of the same magnitude as the observed trends in the 20th century.
- Station histories are rarely complete and, therefore, do not alone provide a reliable basis for the evaluation of homogeneity. Statistical homogeneity testing and adjustment provide useful tools for the production of reliable data sets, but they may not be sufficient. The best result will be achieved by using both statistical tests and metadata information.

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